

Impact of control techniques for conducted emission through modelling and simulation of a BLDC drive

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Abstract. This paper deals with the modelling approach and simulation of conducted emission in brushless permanent magnet DC drives with various control techniques. The model incorporates CISPR 25-based conducted emission measurement setup, discrete component behaviour, high-frequency motor models, and PCB parasitics. Hysteresis current control, sinusoidal and space vector pulse width modulation techniques are used to control the switching elements and compared based on their conducted emission performance.

1 Introduction

The increasing integration of power electronics into modern systems reflects a global shift toward decarbonization and sustainable energy solutions. As industries transition to higher shares of electric power, the role of power electronics, especially switching transistors, becomes critical. Recent advances in transistor technologies, including silicon-carbide (SiC) and gallium-nitride (GaN) devices, have enabled higher switching frequencies, significantly improving performance by reducing heat dissipation and decreasing system size. However, these technological advancements come with challenges, particularly the generation of higher electromagnetic emissions due to faster switching. For instance, a tenfold increase in rise time can lead to a 20 dB increase in emissions at high frequencies, creating design and compliance challenges for engineers [1].

Electromagnetic compatibility (EMC) issues in power electronic systems, such as conducted and radiated electromagnetic interference (EMI), have become prominent concerns. EMC ensures that devices operate as intended without causing or being affected by electromagnetic disturbances. The need for accurate simulation tools has grown as designers aim to predict and mitigate EMI during the design phase, reducing the need for costly redesigns. This paper contributes to this field by presenting an advanced methodology for modelling and predicting EMI in power electronic systems.

Recent studies have explored diverse methodologies to simulate and model conducted emissions (CE) in power electronics. For instance, Lu and Guru [2] modelled a servo drive system using a high-frequency model of permanent magnet synchronous motor and achieving accurate predictions of conducted emissions in low and high frequencies by impedance-measurement-based XY capacitors and common mode choke, circuit model of long cable with extracted parasitics of heat sink and power and ground nets. Similarly, Bodake et. al. [3] applied a numerical simulation-based parasitic model of the printed circuit board (PCB), observing comparable accuracy between simulation and measurement results for a broad frequency range. These studies highlight the importance of accurate parasitic modelling for efficiently conducted emission predictions. [4] highlights the relevance of the corresponding modelling of load and its parasitic parameters. In addition to analytical calculations, the author also uses circuit simulations to demonstrate the effect of load parasitics on the conducted emission.

In contrast, Riener et al. [5] utilized 3D finite element (FE) simulations to develop circuit models for the half-bridge stage, focusing on the methodology of the simulation and extending the frequency range up to 400 MHz. In this work, the authors show how to implement and set up a simulation for a power converter with a simplified transistor model and a vector-fitted S-parameter

model. While this approach demonstrated the accuracy is within 12 dB, limitations emerged in some cases, underscoring the need for a more comprehensive transistor model. [6] also provides a comprehensive overview of CE modelling in power electronics. There is no specific simulation here, but the article does an excellent job of summarizing the literature on the subject.

The commonly used control techniques play an important role in the development of electromagnetic emissions. The applied technique determines the harmonics associated with the switching frequency up to the MHz range, while the large time variation of voltage and current creates harmonics in the MHz range [7]. Even though several studies have addressed the comparison of modulation techniques in drive systems [7]–[9]. Jiang et al. [7] provide insights on common-mode noise reduction of three-phase inverters for different modified pulse width modulation (PWM) methods. In this case, the modulation techniques are not tested on a real drive system. For the other articles [8], [9], the authors investigate the drive system and compare the control technique, but based on classical drive properties.

The literature review found no studies assessing conducted emission from the comparison of different control techniques of modulation for electric drive systems. Therefore, the objective of this paper is to present the methodology of the CE simulation using a brushless direct current (BLDC) motor drive. In addition, to compare the most commonly used control methods in the automotive industry from the CE point of view using a simulation model of the BLDC drive.

2 Conducted emission modelling

The analysed BLDC drive is used in an automotive application and thus has to comply with the CISPR 25 standard [10] for EMC. Hence, the simulation model should be built according to the arrangement specified in the standard. In the following, the main parts of the simulation model will be briefly reviewed, highlighting the parts where it differs from similar work [2], [3], [5]. The complete simulation model was assembled in Ansys Circuit [11], as shown in Fig. 1.

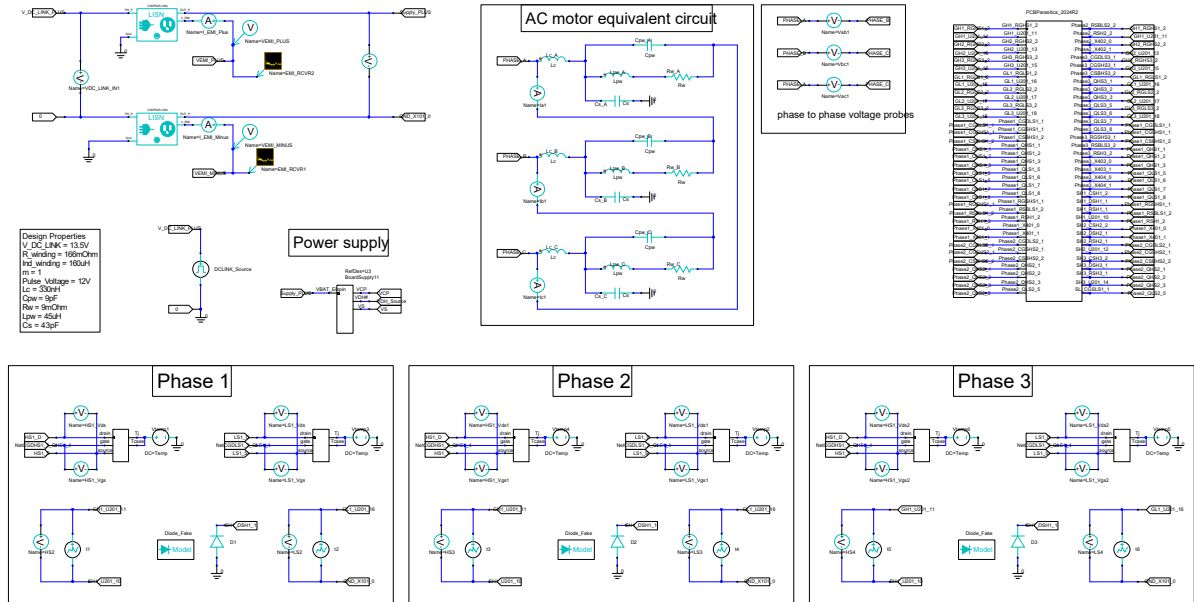


Figure 1: Circuit model of the BLDC motor drive with S-parameter model of PCB.

2.1 LISN

The line impedance stabilization network (LISN) [1] is an essential part of the test because it ensures the reproducibility of the measurement. Here, the LISN required by the CISPR 25 standard was used in the circuit, as shown in [3]. What differs from the layout shown in [3] is the use of the built-in EMI receiver block. This block simplifies the evaluation of transient signals as required by the standard with the desired type of detector.

The impedance of the LISN model was compared with the impedance of a genuine artificial mains network (*Schwarzbeck Mess NNB M 8124*) and found a good match in the 1 kHz - 100 MHz range.

2.2 Numerical model of conducted emission setup with PCB

The main part of the simulation model is built similarly as presented in [3], [5]. For the progress of the results as close as possible to reality, it is necessary to bear in mind cables, ground plane, and low relative permittivity support in addition to PCB as shown on the left side of Fig. 2.

The best solution is to create an S-parameter model from the 3D numerical one and use it in the circuit simulation of the conducted emission. In this case, the simulation based on the 3D finite element method [12] was performed in the Ansys HFSS [11] environment. The creation of the S-parameter model, the wave equation prescribed for the electric field strength solved in the frequency domain with the proper boundary conditions [12],

$$\nabla \times \left(\frac{1}{\mu} \nabla \times \vec{E} \right) - \omega^2 \varepsilon \vec{E} = -j\omega \vec{J}_{\text{imp}}, \quad (1)$$

where $\vec{E}$ is the electric field strength, $\mu = \mu_0 \mu_r$ and $\varepsilon = \varepsilon_0 \varepsilon_r$ are the permeability and permittivity, respectively, ω is the angular frequency, $\vec{J}_{\text{imp}}$ is the prescribed current density.

What's new in this area is that the 3D CE environment includes the PCB as a so-called layout component, which allows for a high degree of scalability in examining complex layouts in a 3D environment. In addition to the layers, the layout component contains the ports and passive components as you see on the right side of Fig. 2. It is beneficial to assemble the model this way because the software can easily handle vast size differences (e.g. support and a pad) using the Ansys Mesh Fusion technique [11]. This way, the environment and the PCB will have their mesh corresponding to the geometry, and another meshing algorithm will be used.

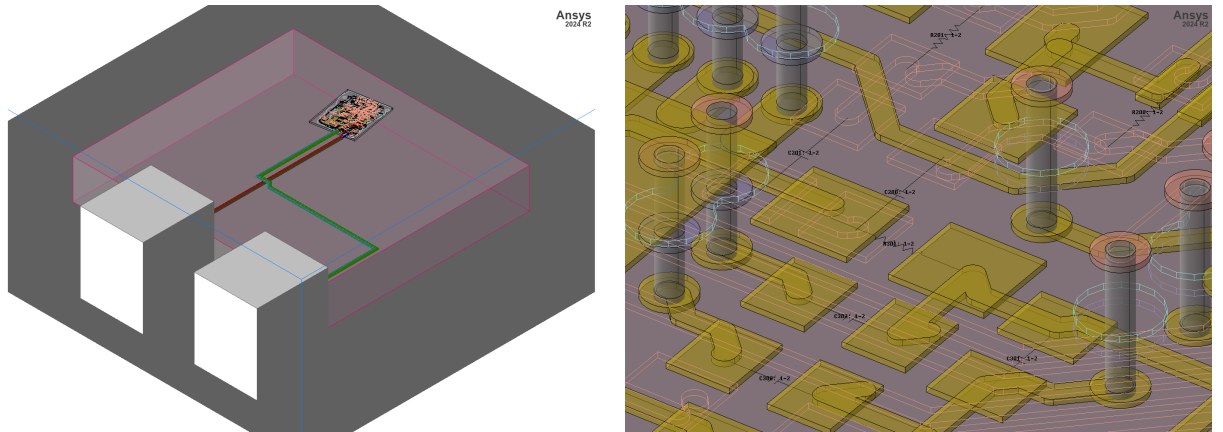


Figure 2: The 3D model of the conducted emission test setup (*left*) and the lumped elements in the PCB model (*right*).

It is important to note that the PCB has ports with different impedances, so it is recommended to export the S-parameter model without renormalization otherwise, the power and ground impedances will not match.

2.3 *Passive and active components*

The passive components (R, L, C) are either idealized lumped elements with ESR and ESL values or as Spice (.lib) models in the layout component. The latter format is used for the more relevant components like DC link capacitances or capacitances and inductances in the filter.

Active components such as diodes and transistors are also included in the simulation in the form of Spice models. Components are connected to the circuit via ports defined in the 3D model. One of the conclusions in [5] is that it is principal to model the transistor as accurately as possible for more realistic results. A separate simulation was required to verify the model accuracy and compare its characteristics with the data sheet curves on different complexity models provided by the transistor manufacturer.

2.4 *Control techniques*

In this work, the most commonly applied control techniques in the automotive industry, the hysteresis current control (HCC), the sinusoidal PWM (SPWM) and the space vector PWM (SVPWM) control techniques [7]–[9] used and compared on the bases of the conducted emission.

In HCC [9], the control compares the inverter output current to a reference value and keeps it within a specified band. It is the proper choice for smaller drives, as it is straightforward in its algorithm and has good dynamic control properties.

In sinusoidal pulse-width modulation (SPWM) [7]–[9], a triangular carrier wave is compared to a sinusoidal reference wave at the desired fundamental frequency, and their relative amplitudes determine the pulse widths, which control the switching of devices in each phase leg of the inverter. SPWM is a very popular method, where the output signal has less harmonic content than the square wave inverter, but high-frequency components appear in the output as a function of the frequency modulation index [9]

The fundamental concept of SVPWM [7]–[9] is to generate a reference voltage vector that closely approximates the ideal voltage vector by utilizing eight distinct voltage vectors produced by the inverter. This method aims to reduce switching losses and achieve greater DC bus utilization compared to SPWM.

The applied control techniques were not directly implemented in the simulation, instead, the control signals for the transistors were generated and used in the simulation. For PWM signals, the carrier frequency is 20 kHz, and the sine frequency is 1 kHz. In all cases, the dead time between switching was 1.5 μ s. For HCC, the band limit was ± 0.2 A relative to the reference current signal.

2.5 *BLDC motor model*

The impedance observed at the LISN influences electromagnetic emissions, which is significantly affected by the parasitic parameters of the load [4]. For example, parasitic capacitances between the load terminals and their chassis, along with the inductance of interconnection cables, can impact emission levels. The contributions of BLDC motor parasitics to the conducted emissions are a high-frequency model of the load needed within the conducted emission frequency range, and the model can be simplified as depicted in Fig. 1. The used motor model differs from the one in [4] because the motor phase resistance is contained in the circuit.

The parameters of the high-frequency motor model were obtained from the motor datasheet and as described in the article [3] for the heat sink using quasi-static 3D field simulation.

3 Results

The circuit simulation of the BLDC drive system was running in the time domain. The correct operation of the system is verified best by looking at the phase currents of the motor. The currents shown in Fig. 3 indicate that the drive model operation is good. For clarity, only the phase A current has been visible in all three cases. Looking at the current waveforms, it is clear that the sinusoidal PWM has the least number of switches. The hysteresis control has the most switching because of the frequent on and off of the transistor due to the band. Increasing the band would reduce the number of switches, but the current ripple would be high.

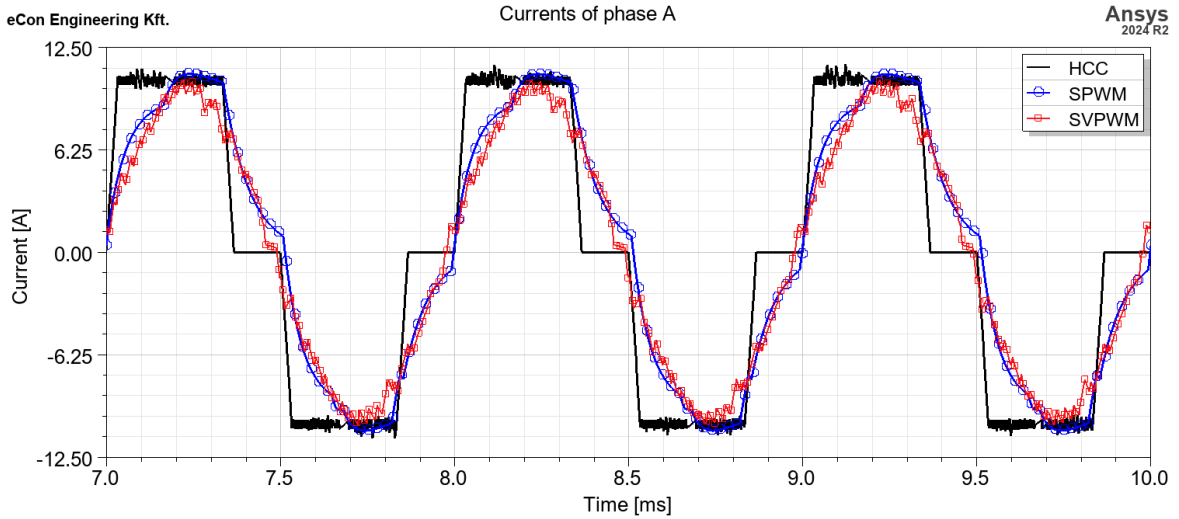


Figure 3: Phase current of the BLDC motor for different control techniques.

The statements for motor phase current are well supported by the results obtained for the conducted emissions. Fig. 4 shows the spectrum of the simulated conducted emission. The curves represent peak values. The limit line refers to the peak value. The simulation results show that the BLDC motor drive system with SPWM meets the requirement of CE standard, i.e. up to 30 MHz, the emission is under the standard limit. SVPWM has higher emissions than SPWM but only exceeds the standard limits above 15 MHz. For HCC, the emission is close to the limits at 2 MHz but does not exceed them here. However, above 3 MHz, the emission is already above the limit.

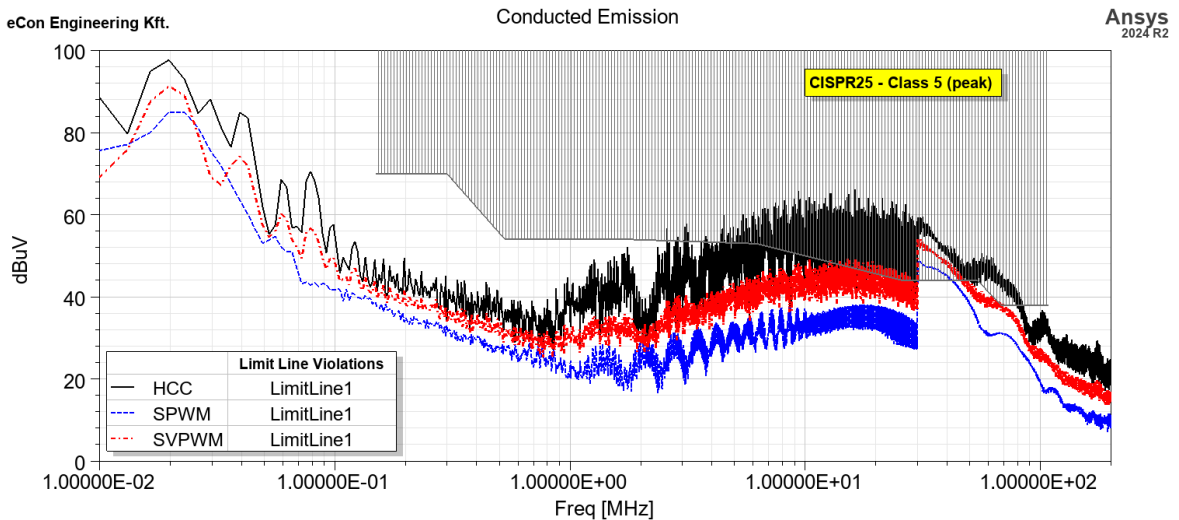


Figure 4: Simulation results of conducted emission with CISPR 25 limit line.

4 Conclusion

A brief review of the connected literature with the methodology used for conducting emission modelling is described in this paper. The presented modelling techniques were used for a BLDC motor drive system with HCC, SPWM and SVPWM control techniques. Due to the capabilities of software and high-performance computing, it is possible to study increasingly complex models in one. However, the most efficient and robust technique remains the connection of a behavioural model obtained by the 3D frequency domain numerical model and time domain circuit simulation. Simulation provides valuable early insights into which parts of the system or specific components contribute to emissions. This approach allows for finding potential issues early in product development, enabling designers to take appropriate measures and make informed recommendations to mitigate emissions.

The results clearly show that the best control technique from the point of view of conducting emissions may not be the best one. Simulation can also be a useful tool here, for different control strategies can be quickly tested, avoiding the need for a retrofit ferrite ring or choke to ensure the system meets the standard.

My further plans are to improve the high-frequency model of the motor and take into account the effects of its housing parasitic in the 3D numerical model. Also, to make the switching of the semiconductors more realistic, including the behaviour of the IC that drives the transistors.

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